

The Relational Model

Databases - Computer Science

1. Relations, tuples and domains

A relation is a set of tuples over a fixed set of attributes. Being a set, it has no inherent row order and no duplicates. Each attribute draws values from a domain, and NULL marks information that is absent or inapplicable.

2. Keys

A candidate key is a minimal attribute set that uniquely identifies a tuple. One is designated the primary key; the rest remain alternate keys, usually enforced with unique constraints. A foreign key holds values that must appear as a key in another relation, which is how referential integrity is expressed.

3. Integrity constraints

Entity integrity forbids a NULL in any part of a primary key. Referential integrity requires every non-null foreign key value to match an existing key. Domain constraints restrict the values an attribute may take. Together they are the guarantees the database enforces regardless of the application on top of it.